

■ EXAMPLE 7.12

In this example, we consider a vehicle navigation problem. The first two state elements are the north and east positions of a land vehicle, and the last two elements are the north and east velocities. The velocity of the vehicle is in the direction of θ , an angle measured clockwise from due east. A position-measuring device provides a noisy measurement of the vehicle's north and east positions. The process and measurement equations for this system can be written as

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 0 \\ T \sin \theta \\ T \cos \theta \end{bmatrix} u_k + w_k \\ y_k &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} x_k + v_k \end{aligned} \quad (7.173)$$

where T is the discretization step size. We can implement a Kalman filter to estimate the position and velocity of the vehicle based on our noisy position measurements. If we know that the vehicle is on a road with a heading of θ , then we know that

$$\begin{aligned} \tan \theta &= x(1)/x(2) \\ &= x(3)/x(4) \end{aligned} \quad (7.174)$$

These constraints can be written as

$$\begin{bmatrix} 1 & -\tan \theta & 0 & 0 \\ 0 & 0 & 1 & -\tan \theta \end{bmatrix} x_k = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (7.175)$$

The constrained filter can be implemented using any of the four approaches discussed in this section (model reduction, perfect measurements, projection, or pdf truncation). Figure 7.4 shows the magnitude of the north position estimation error of the unconstrained and constrained filters (projection approach using $W = I$) for a typical simulation. In this example, significant estimation improvement can be obtained when constraint information is incorporated into the filter, although the improvement will be problem dependent.

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It is clear from this section that there are a variety of ways to enforce equality or inequality constraints on state estimation problems. The “best” way is not clear-cut, and probably depends on the application. Other approaches to constrained estimation and some discussion of the mathematical meaning of state constraints can be found in [Hel94, Rao03, Dew04, Goo05a, Goo05b, Ko06].

7.6 SUMMARY

In this chapter, we discussed a variety of Kalman filter generalizations that make the filter more widely applicable to a broader class of problems. Correlated and colored process and measurement noise were studied early in the history of the

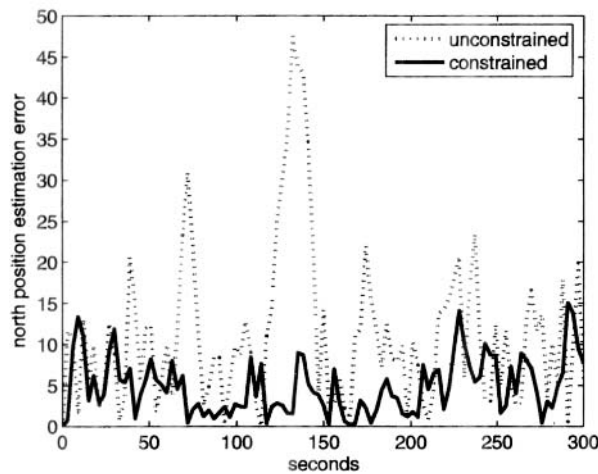


Figure 7.4 North position estimation error magnitude of the unconstrained and constrained Kalman filters for Example 7.12.

Kalman filter. We showed in this chapter that filter modifications taking correlation and color into account can improve estimation performance. However, whether or not these approaches are worth the extra complexity and computational effort is problem dependent. One of the most practical extensions of the Kalman filter is the steady-state Kalman filter. The steady-state Kalman filter often performs nearly identically to the more theoretically rigorous time-varying filter. However, the steady-state filter requires only a fraction of the computational cost. The α - β and α - β - γ filters are special cases of the steady-state Kalman filter. We also discussed the fading-memory filter, which is a way of making the Kalman filter more robust to modeling errors. The fading-memory filter is a simple modification to the Kalman filter that can noticeably improve filter performance. Further discussion of filter robustness is found in Section 10.4 and Chapter 11. Finally, we discussed several ways to incorporate state constraints in the Kalman filter to improve estimation accuracy when information other than the state model is available. Other Kalman filter generalizations are discussed in later chapters of this book.

- Kalman filters with fewer states than the system (Section 10.3)
- Kalman filtering when the system model or noise statistics are not known (Section 10.4)
- Kalman filtering when the measurements arrive at the filter in the wrong order (Section 10.5)
- Kalman filters for nonlinear systems (Chapter 13)

Further generalizations undoubtedly await future development by the efforts of enterprising students and researchers.

PROBLEMS

Written exercises

7.1 Consider the scalar system

$$\begin{aligned}x_k &= \frac{1}{2}x_{k-1} + w_{k-1} \\ y_k &= x_k + v_k \\ v_k &= \frac{1}{2}v_{k-1} + \zeta_{k-1}\end{aligned}$$

where $w_k \sim (0, Q)$ and $\zeta_k \sim (0, Q_\zeta)$. Let $Q = Q_\zeta = 1$.

- Design a Kalman filter in which the dynamics of the measurement noise v_k are ignored and it is assumed that v_k is white noise with a variance of Q_ζ . Based on the incorrect Kalman filter equations, what does the Kalman filter think that the steady-state *a posteriori* estimation covariance is?
- Based on the incorrect Kalman filter equations, what is the true steady-state *a posteriori* estimation covariance $E(e_k^2)$? Hint: Find a recursive equation for $E(e_k^2)$ in terms of $E(e_{k-1}^2)$, $E(w_{k-1}^2)$, $E(v_k^2)$, and $E(e_{k-1}v_k)$, then solve for the steady-state value of $E(e_k^2)$.
- Design a Kalman filter using the state augmentation approach in which the dynamics of the measurement noise are correctly taken into account. What is the steady-state estimation covariance? Hint: You may need to use MATLAB's DARE function to solve the steady-state Riccati equation that is associated with this question.

7.2 Show that the Kalman filter for an LTI system with a noise-free scalar measurement that satisfies the equation $(HQH^T)Q = QH^THQ$ has a steady-state *a posteriori* covariance of zero.

7.3 Consider the scalar system

$$\begin{aligned}x_k &= x_{k-1} + w_{k-1} \\ y_k &= x_k + v_k\end{aligned}$$

where $w_k \sim (0, Q)$ and $v_k \sim (0, R)$ are white noise processes with $Q = R = 1$. Suppose that $E(w_kv_{k+1}) = M = 1$.

- Design a Kalman filter in which the correlation between w_k and v_{k+1} is ignored. Based on the incorrect Kalman filter equations, what does it appear that the steady-state *a posteriori* estimation covariance is?
- For the Kalman filter designed above, write a recursive equation for the *a posteriori* estimation error $e_k = x_k - \hat{x}_k^+$. Use this equation to find the steady-state solution to $E(e_k^2)$.
- Design a Kalman filter in which the correlation between w_k and v_{k+1} is correctly taken into account. Show that the steady-state *a posteriori* estimation covariance is zero. Explain why the estimation covariance goes to zero in spite of the existence of process noise and measurement noise. (Hint: Use the correlation between w_k and v_{k+1} to write an equivalent two-state system, and then use the results of Problem 7.2.)

7.4 Consider the system

$$\begin{aligned}x_k &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} x_{k-1} + w_{k-1} \\ y_k &= \begin{bmatrix} 1 & 1 \end{bmatrix} x_k + v_k\end{aligned}$$

where $w_k \sim (0, Q)$ and $Q = I$.

- Find one matrix square root of Q .
- Is (F, H) observable?
- Is (F, H) detectable?
- Is (F, G) controllable for all G such that $GG^T = Q$?
- Is (F, G) stabilizable for all G such that $GG^T = Q$?
- Use the above results to specify how many positive definite solutions exist to the DARE that is associated with the Kalman filter for this problem.
- Use the above results to specify whether or not the steady-state Kalman filter for this system is stable.

7.5 Prove that the matrix $\mathcal{H}$ in Equation (7.85) is symplectic.

7.6 In this problem, we will use the shorthand notation $P = P^+$ and $M = P^-$. Use the following procedure to find α as a function of β for the α - β filter [Bar01].

- Use the time-update equation for M to solve for the three unique elements of P as a function of the three unique elements of M .
- Use the measurement-update equation for P to solve for the three unique elements of P as a function of the three unique elements of M .
- Equate the sets of equations from the two steps above to get expressions for $M_{11}K_1$, $M_{12}K_1$, and $M_{12}K_2$, that do not have any P_{ij} terms.
- Use Equation (7.64) to solve for M_{11} and M_{12} .
- Combine the five equations from the two previous steps to get a single equation with K_1 and K_2 that does not have any M_{ij} terms.
- Replace K_1 and K_2 in the previously obtained equation with α and β from Equation (7.63), then solve for α as a function of β .

7.7 Prove the properties of symplectic matrices that are listed immediately following Equation (7.86).

7.8 Recall that the steady-state, zero-input, one-step formulation of the *a posteriori* Kalman filter can be written as

$$\begin{aligned}\hat{x}_k^+ &= (I - KH)F\hat{x}_{k-1}^+ + Ky_k \\ \hat{y}_k &= H\hat{x}_k^+\end{aligned}$$

Prove that if (F, H) is observable and $(I - HK)$ is full rank, then the Kalman filter in the above equation is an observable system. Hint: $H(I - KH) = (I - HK)H$.

7.9 Suppose you have a two-state Newtonian system of the type described in Section 7.3.1. The sample time is 1 and the variance of the acceleration noise is 1. A requirement is given to estimate the position with an *a posteriori* steady-state variance of 1 or less. What is the largest measurement variance that will meet the requirement?

Computer exercises

7.10 Consider the system described in Problem 7.1. Implement the Kalman filter that assumes white noise and the Kalman filter that assumes colored noise. Numerically calculate the RMS *a posteriori* estimation-error variance and verify that it matches the analytically calculated values from your answer to Problem 7.1.

7.11 Plot the α and β parameters of the α - β filter as a function of λ . Use a log scale for λ with a range of 10^{-3} to 10^3 . What are the limiting values of α and β as $\lambda \rightarrow 0$? Does this make intuitive sense? What are the limiting values of α and β as $\lambda \rightarrow \infty$?

7.12 Plot the α , β , and γ parameters of the α - β - γ filter as a function of λ . Use a log scale for λ with a range of 10^{-3} to 10^3 . What are the limiting values of α , β , and γ as $\lambda \rightarrow 0$? Does this make intuitive sense? What are the limiting values of α , β , and γ as $\lambda \rightarrow \infty$?

7.13 A simple model of the ingestion and metabolism of a drug is given as

$$\begin{aligned}\dot{x}_1 &= -k_1 x_1 + u \\ \dot{x}_2 &= k_1 x_1 - k_2 x_2 \\ y(t_k) &= x_2(t_k) + v(t_k)\end{aligned}$$

where the units of time are days, x_1 is the mass of the drug in the gastrointestinal tract, x_2 is the mass of the drug in the bloodstream, and u is the ingestion rate of the drug. Suppose that $k_1 = k_2 = 1$. The measurement noise $v(t_k)$ is zero-mean and unity variance. The initial state, estimate, and covariance are

$$\begin{aligned}x(0) &= \begin{bmatrix} 0.8 \\ 0 \end{bmatrix} \\ \hat{x}(0) &= x(0) \\ P(0) &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\end{aligned}$$

It is known from physical constraints that $x_1 \in [0.8, 1]$.

- a) Discretize the system with a step size of 1 hour.
- b) Implement the discrete-time Kalman filter, the projection-based constrained Kalman filter with $W = I$, and the pdf truncation constrained filter. Run simulations of these filters for a three-day period. Plot the magnitude of the x_1 estimation error for the three filters. Which filter appears to perform best? Which filter appears to perform worst?

